



Enhancing Event-Frame Feature Tracking with Siamese FPN

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Motivation

Current learning-based event-frame trackers typically encode the reference frame and event data using separate networks. This introduces redundant parameters and does not explicitly constrain both modalities to produce compatible features.

Contributions

- The replacement of the separate encoders with a shared-weight Siamese FPN for aligned event-frame feature extraction.
- A RAFT-style loss supervising all intermediate tracking updates for improving robustness under synthetic-to-real domain shift.

Benefits

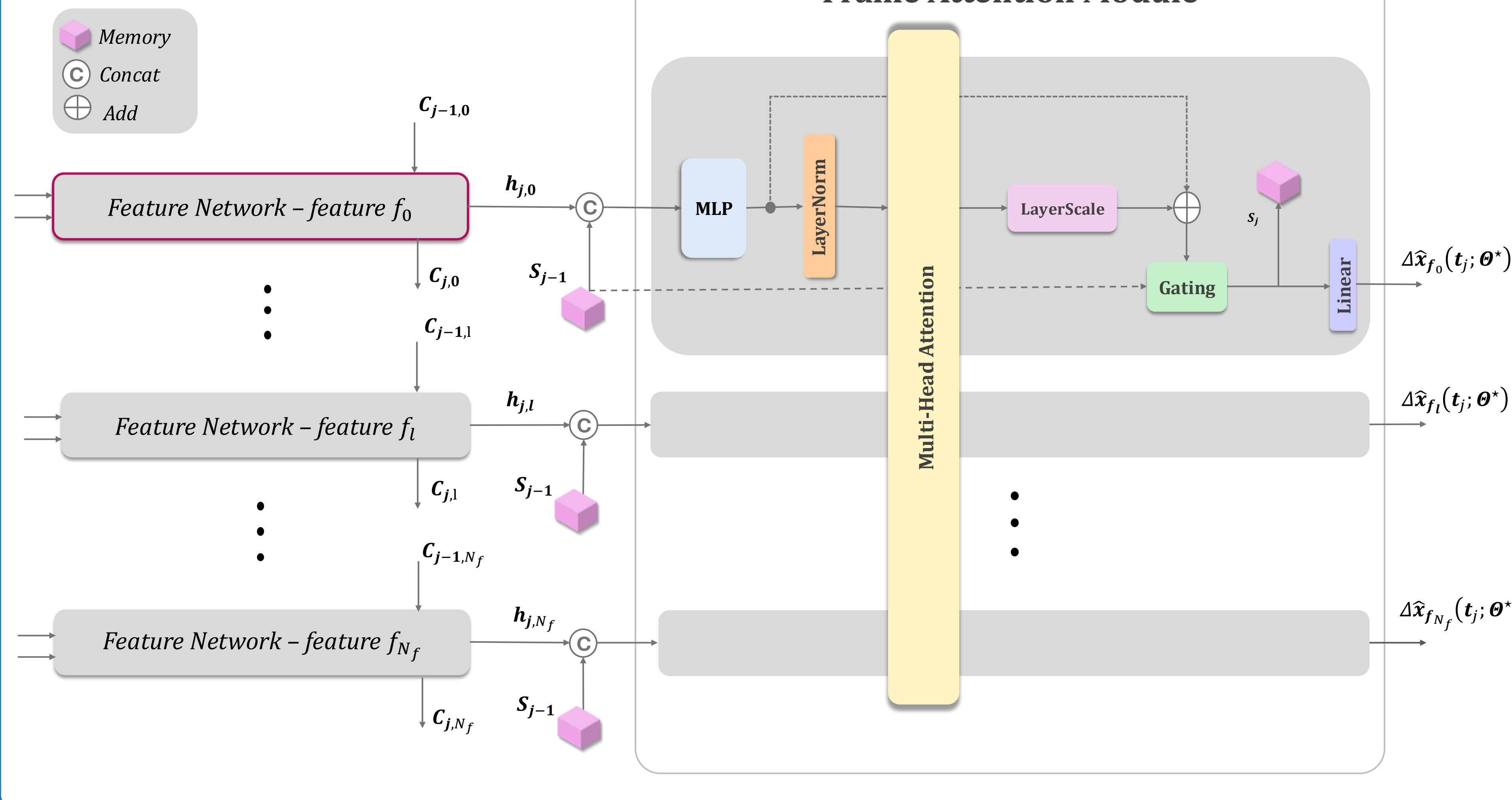
- The proposed model:
- reduces the parameters from approximately 35M to 20M
- significantly lowers GPU memory
- maintains comparable FPS, latency, FLOPs and
- Improves performance

Method Overview

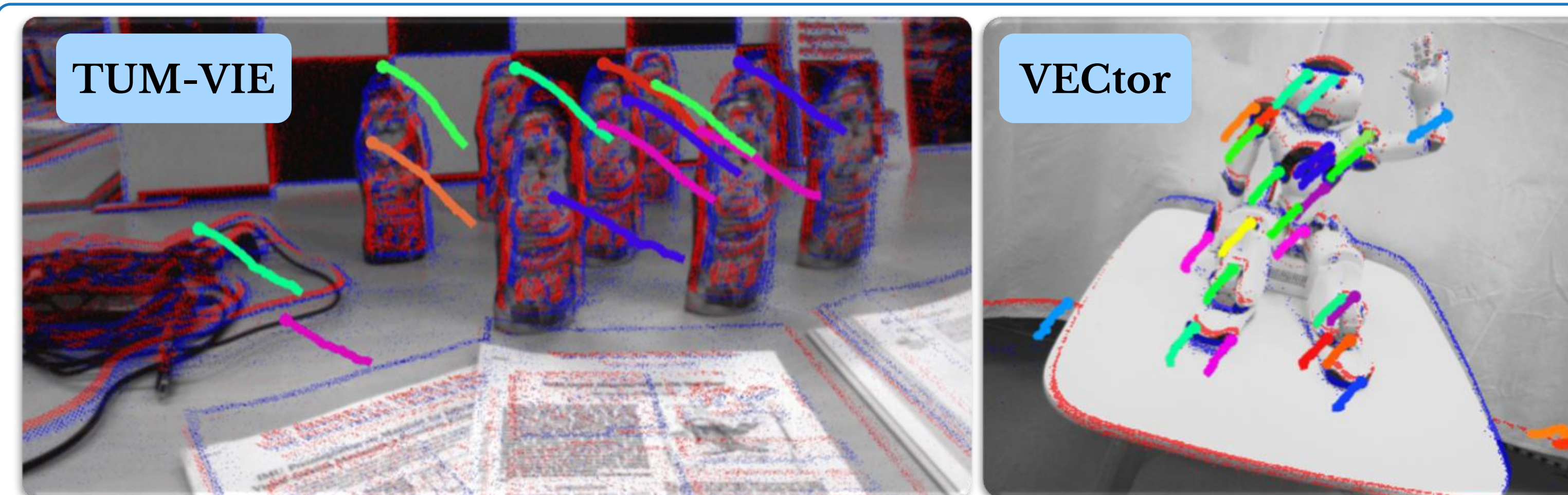
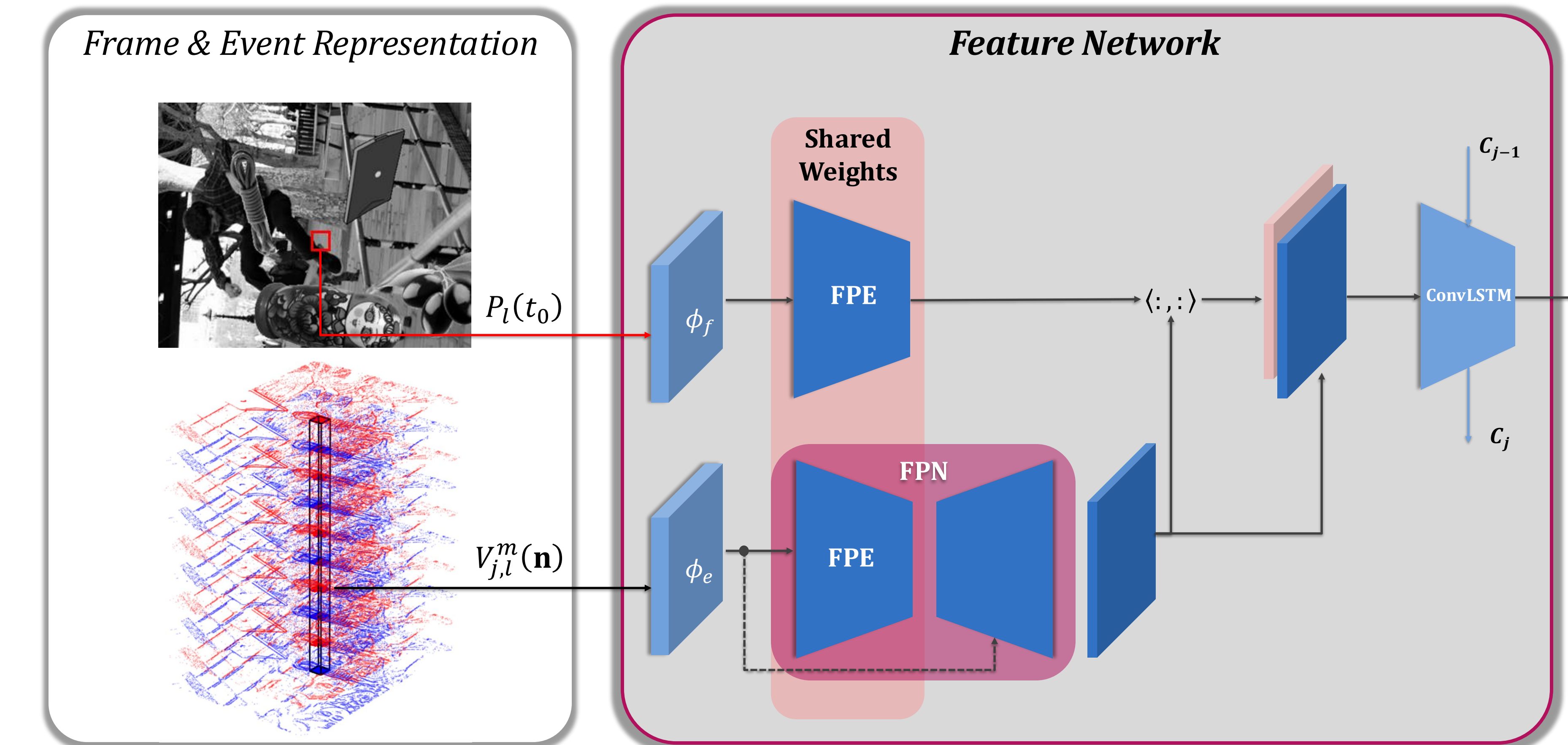
- For each initialized feature, a reference patch is extracted from the initial frame and kept fixed throughout tracking.
- At every tracking step:
- Events are cropped around the current feature estimate.
- Frame and event inputs pass through modality-specific stems.
- A shared FPE extracts multi-scale descriptors.
- The frame descriptor is correlated with the event feature map.
- A ConvLSTM and Frame Attention Module refine the match.
- The network predicts the next incremental displacement.
- The feature position is updated according to:

$$\hat{x}_{f_l}(t_j) = \hat{x}_{f_l}(t_{j-1}) + \Delta \hat{x}_{f_l}(t_j).$$
- The updated position determines the event search region for the next step.

System Architecture



Siamese Shared-Weight FPN

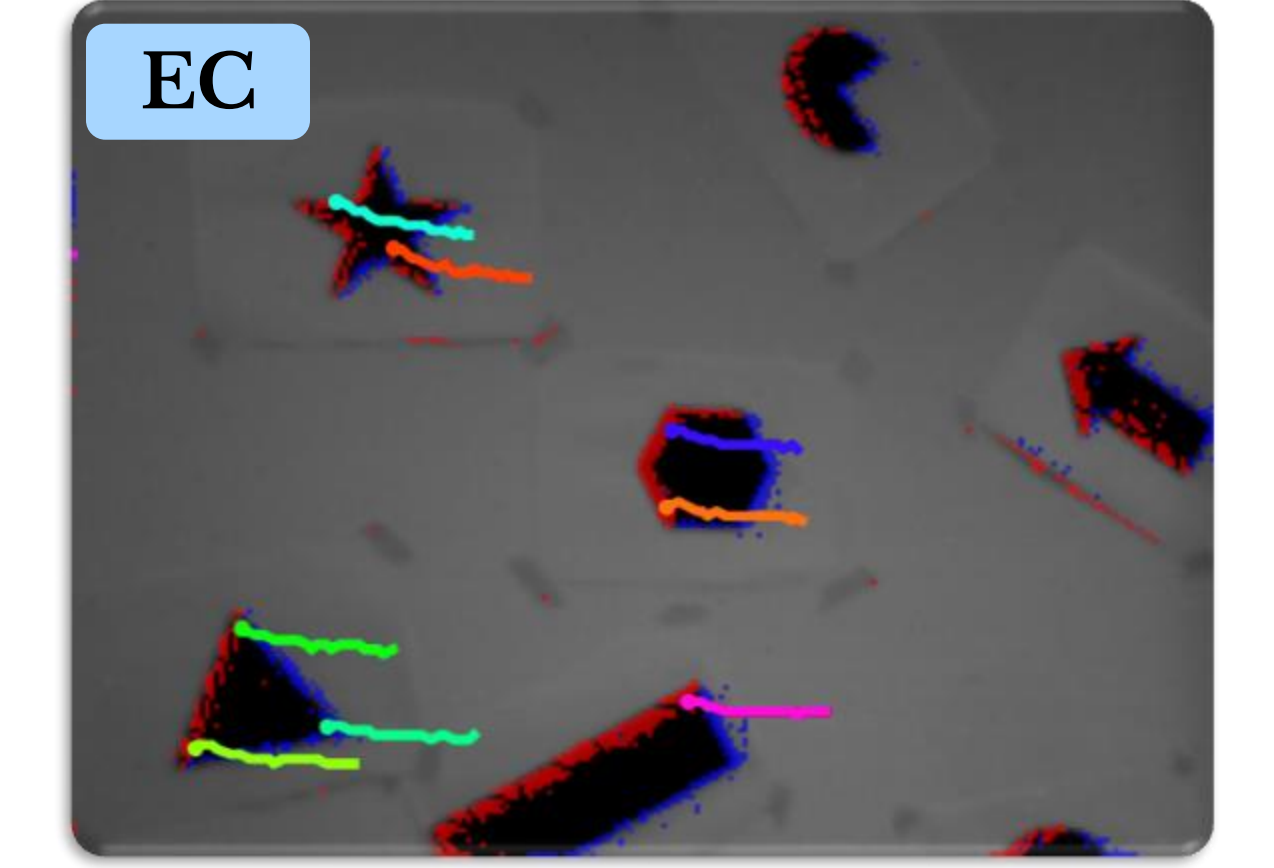
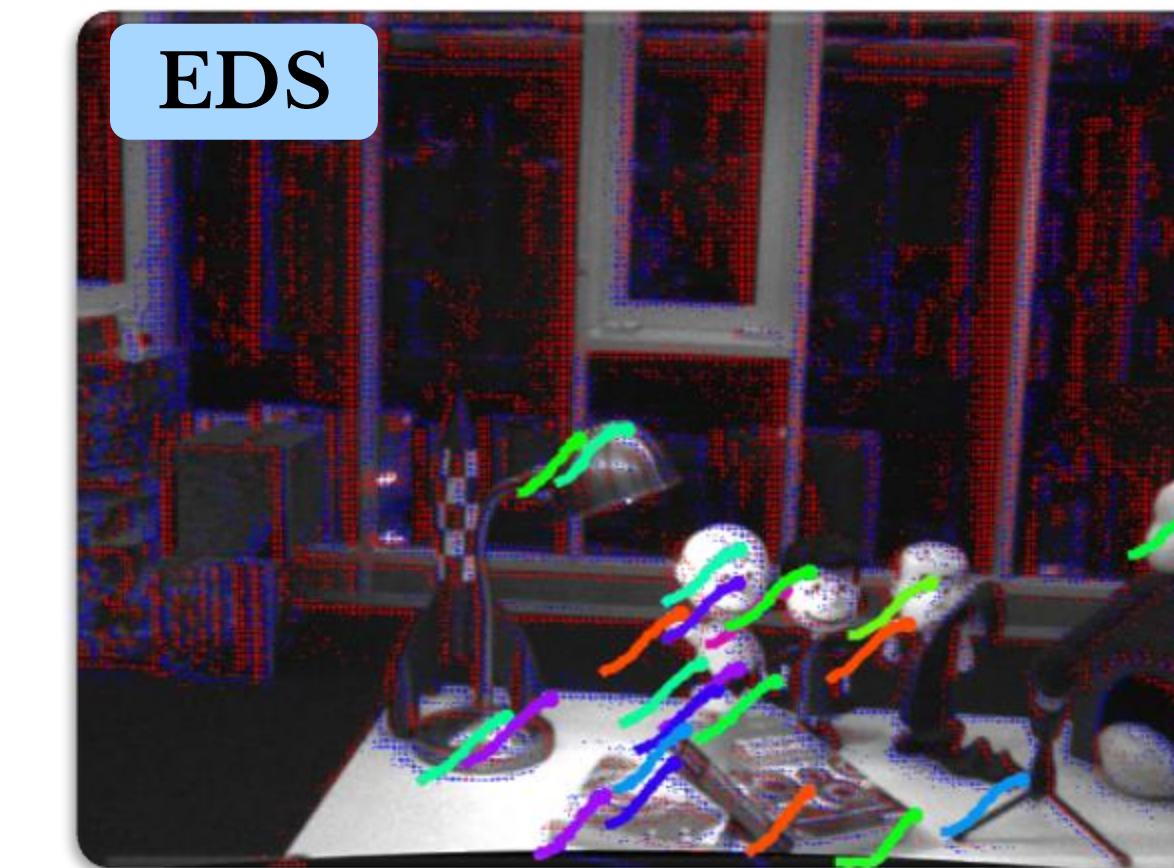


Benchmarking Results

The proposed pipeline substantially improves over DDFT on both real datasets and achieves the best event-frame performance on EC.

Method	Input	EDS		EC	
		FA ↑	EFA ↑	FA ↑	EFA ↑
ICP (Kung et al.)	E+F	0.060	0.040	0.256	0.245
EM-ICP (Zhu et al.)	E	0.161	0.120	0.337	0.334
HASTE (Alzugaray et al.)	E	0.096	0.063	0.442	0.427
EKLT (Gehrig et al.)	E+F	0.325	0.205	0.811	0.775
DDFT (zero-shot) (Messikommer et al.)	E+F	0.549	0.451	0.795	0.787
DDFT (fine-tuned) (Messikommer et al.)	E+F	0.576	0.472	0.825	0.818
FE-TAP (Liu et al.)	E+F	0.676	0.589	0.844	0.838
Ours ($\gamma = 0.9$)	E+F	0.616	0.521	0.866	0.858
ETAP (Hamann et al.)	E	0.701	0.601	0.881	0.876
MATE (Han et al.)	E	0.713	0.626	0.885	0.875

The most recent event-only methods still obtain the highest overall scores, especially on EDS.



Ablation Results

Method	Components				EDS		EC	
	FAM	Siamese	Conv-128	Loss	FA ↑	EFA ↑	FA ↑	EFA ↑
DDFT	✗	✗	✗	Truncated	0.549	0.451	0.795	0.787
DDFT	✗	✗	✗	RAFT	0.506	0.415	0.827	0.820
DDFT + FAM	✓	✗	✗	Truncated	0.614	0.500	0.832	0.825
Siamese DDFT	✗	✓	✗	Truncated	0.507	0.427	0.781	0.775
Siamese DDFT + Conv-128	✗	✓	✓	Truncated	0.575	0.482	0.820	0.802
Siamese DDFT + Conv-128 + FAM	✓	✓	✓	Truncated	0.604	0.515	0.854	0.843
Siamese DDFT + Conv-128 + FAM	✓	✓	✓	RAFT	0.616	0.521	0.866	0.858

Conclusion

- The proposed Siamese FPN improves synthetic-to-real generalization by sharing the multi-scale encoder between frame and event branches.
- Although the RAFT-style loss provides little benefit on MultiFlow, it consistently improves performance on the real EDS and EC datasets by supervising the full unrolled sequence.
- These results show that shared event-frame encoding and sequence-level supervision enable more robust real-world feature tracking.